

SIGNIFICANCE TESTS

master recipe • also used in Units 4 and 5

ZONE A • THE FORMULAS + THE FOUR-STEP FRAME

$$z = \frac{\text{statistic} - \text{parameter under } H_0}{\text{standard error of the statistic}}$$

SE estimates the statistic's sampling SD; use the test's null model. A t -procedure uses t with its **df**.

$$P\text{-value} = P(\text{observed or more extreme statistic} \mid H_0 \text{ true})$$

THE SAME FOUR STEPS AS A CONFIDENCE INTERVAL

- 1 STATE** Define the parameter. Give H_0 , H_a , and α in context.
- 2 PLAN** Name the test. Check Random $\rightarrow$ 10% $\rightarrow$ Normal / Large Counts.
- 3 DO** Statistic, df if needed, and P-value. Draw and shade the curve.
- 4 CONCLUDE** Compare P with α ; state evidence for H_a in context.

THE DECISION

$P \leq \alpha$
 $\rightarrow$ reject H_0
Convincing evidence for H_a .

$P > \alpha$
 $\rightarrow$ fail to reject H_0
Not convincing evidence for H_a .

ZONE B • ERRORS + POWER

DECISION $\times$ TRUE STATE

	H_0 true	H_a true
Reject H_0	Type I error	Correct
Fail to reject H_0	Correct	Type II error

α : significance level.

β and power depend on a specific true effect.

$$\begin{aligned}\alpha &= P(\text{Type I error}) \\ &= P(\text{reject } H_0 \mid H_0 \text{ true})\end{aligned}$$

$$\begin{aligned}\beta &= P(\text{Type II error}) \\ &= P(\text{fail to reject } H_0 \mid H_a \text{ true})\end{aligned}$$

$$\text{Power} = 1 - \beta$$

Increases with larger n , larger α , or a larger true effect, with other factors held fixed.

ZONE C • SAY IT IN WORDS

"Because the P-value of _____ is [less/greater] than $\alpha = \underline{\hspace{1cm}}$, we [reject / fail to reject] H_0 . There [is / is not] convincing evidence that _____."

Fill the last blank with H_a in context. At $P = \alpha$, use "equal to" and reject.

ZONE D • WATCH OUT

KEEP THE DECISION PRECISE

Say "fail to reject H_0 ."
 Never "accept H_0 ."

NAME THE PARAMETER

Hypotheses use p or μ ,
 not $\hat{p}$ or $\bar{x}$.

CHOOSE TAILS BEFORE DATA

One-sided vs two-sided changes P. Let the question choose the tail.